

INDIA'S EMBEDDED SYSTEMS MARKET

Get an overview of the rapidly evolving embedded systems market in India

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Globally, the embedded market is evolving rapidly with systems becoming more sophisticated and software workloads increasing in computational size as well as complexity. Even today, consumer electronics design continues to drive the need for higher-performance, lower-power processors delivered to the market in the shortest possible time. But there still exists tremendous pressure to introduce challenging designs resulting in demand for predictable design flows.

Previously, the embedded systems discipline was confined only to the telecom space but today extensive use of microprocessors, microelectronics and sensors within industries such as automotive, aerospace, energy and manufacturing has thrown up immense

potential for embedded products. The market expansion and heightened future potential are fostering the demand for smart technologies and electronic products.

To facilitate this growth, Indian design organisations are being encouraged to move beyond the simple labour cost arbitrage and become true contributors to product innovation. A number of multinationals have already invested in new design entities across India or expanded their existing facilities, making the country a leading contender in the race to become the hub for embedded technologies and solutions.

Embedded software: Next sunrise sector

One of the key drivers of Indian R&D offshoring is the embedded systems market. Many of the top products driving this growth, such as telecom equip-

ment, monitors, meters and microprocessors, include embedded systems. Clearly, embedded software is set to become India's next sunrise sector and there exists a huge possibility for revenue realisation that remains to be tapped. Moreover, a ten-fold increase in talent by 2015—from a current 60,000 professionals to over 600,000—is expected to facilitate growth in the embedded systems industry. This high domestic demand provides a strong argument for key stakeholders to drive the evolution of this industry.

India's capabilities in building embedded systems

In lieu with the market's ubiquitous nature and perpetual demand, the Indian technology industry plays a pivotal role in meeting constant requirements for computing devices considering its strengths lie in its excellent design talent pool. In fact, larger Indian companies are now heading toward higher-end development and increasing their expertise in specific domains, a study revealed. Embedded software developers are also moving to product ownership and away from modular tasks.

With its core competencies in software services, design and R&D, India is at an advantage as customised applications can be built using embedded technology and products can be made user-friendly even for non-IT people. Moreover, emerging opportunities within this space have further cemented Indian software and hardware firms' potential to leverage their capabilities in building embedded

